

14.11.2025 (E)


SOMAIYA
VIDYAVIHAR UNIVERSITY

Semester: July 2025– November 2025		
Maximum Marks: 50	Examination: End-Semester Examination	Duration: 2 Hrs.
Programme code:03		Class: TY Semester: V (SVU2023)
Programme: Electronics & Telecommunication Engineering		
Institute: K. J. Somaiya School of Engineering		Name of the Department: EXTC
Course Code: 216U03C501	Name of the Course: Digital Communication	
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question Statement	Max. Marks																
Q.1	Attempt any two	10																
i)	Explain Sampling theorem for low pass signal.																	
ii)	Explain Convolutional coding.																	
iii)	Explain Intersymbol interference (ISI) with proper diagram.																	
Q.2	Attempt any one	10																
i)	Apply Huffman coding procedure for the following message ensemble. Find average number of bits per message, average information per message interval and efficiency. <table border="1" style="margin: 10px auto; border-collapse: collapse;"> <tr> <td>X</td> <td>X₁</td> <td>X₂</td> <td>X₃</td> <td>X₄</td> <td>X₅</td> <td>X₆</td> <td>X₇</td> </tr> <tr> <td>P</td> <td>0.25</td> <td>0.125</td> <td>0.0625</td> <td>0.0625</td> <td>0.0625</td> <td>0.25</td> <td>0.1875</td> </tr> </table>	X	X ₁	X ₂	X ₃	X ₄	X ₅	X ₆	X ₇	P	0.25	0.125	0.0625	0.0625	0.0625	0.25	0.1875	
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P	0.25	0.125	0.0625	0.0625	0.0625	0.25	0.1875											
ii)	Apply Shannon Fano source coding. Find average number of bits per message, average information per message interval and efficiency. <table border="1" style="margin: 10px auto; border-collapse: collapse;"> <tr> <td>X</td> <td>X₁</td> <td>X₂</td> <td>X₃</td> <td>X₄</td> <td>X₅</td> <td>X₆</td> <td>X₇</td> </tr> <tr> <td>P</td> <td>0.25</td> <td>0.125</td> <td>0.0625</td> <td>0.0625</td> <td>0.0625</td> <td>0.25</td> <td>0.1875</td> </tr> </table>	X	X ₁	X ₂	X ₃	X ₄	X ₅	X ₆	X ₇	P	0.25	0.125	0.0625	0.0625	0.0625	0.25	0.1875	
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Q.3	Attempt any one	10																
i)	Show that duo-binary signalling suffers from error propagation while precoded duo-binary signalling does not.																	
ii)	What is the purpose of eye diagram? Explain the various parameters observed from the eye pattern using diagram.																	
Q.4	Attempt the following	10																
	Explain QPSK receiver with properly labelled diagram and waveform.																	
Q.5	Attempt the following	10																
	An error correcting code has parity check matrix as $\begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}$ - Determine generator matrix G - Decode the received code word 1 1 0 1 1 0. Is this valid code vector?																	